

KENYA MOBILE MONEY & AGENT NETWORK ANALYTICS

Business Insights & Recommendations

Author: Benard Omoga

Project Type: Portfolio Analytics Project (Synthetic Data)

Tools: Python, MySQL, Power BI, DAX

Dataset: 1,000,000 transactions | 100,000 customers | 10,000 agents | 47 Kenyan counties

1. Business Problem

1.1 Context

Kenya's mobile-money agent network has grown rapidly. CBK reported 381,116 active agents in 2024, rising to 473,536 by December 2025, processing roughly KSh 722.5 billion in monthly transaction value. But growth in agent numbers has not translated into growth in agent economics: average annual M-Pesa agent commissions fell to approximately KSh 112,244 for the year ended March 2026. More agents are splitting a commission pool that is not growing proportionally.

(All figures above are public CBK/industry statistics used only to ground realistic assumptions. All transaction-level data in this project is synthetic and does not represent Safaricom, Airtel, or any bank's actual data.)

1.2 Business Problem Statement

You are a Data Analyst at a Kenyan financial-services company operating a large mobile-money agent network. Management has observed that the agent network is expanding, but is unsure whether that expansion is translating into proportional revenue growth, and where operational or risk issues may be eroding profitability.

Central question: How profitable is our agent network, where are we performing well, and where are we losing money or facing operational risk?

This project answers that question using a synthetic but business-realistic dataset, a dimensional data model, and a Power BI report built on SQL-backed data and DAX measures.

1.3 Data Architecture

To answer this question reliably, a full data pipeline was built from a dimensional (star schema) database design through synthetic data generation to an interactive Power BI dashboard, modeling realistic mobile money transaction patterns across Kenya's 47 counties, three providers (M-Pesa, Airtel Money, Bank Agency Banking), and nine transaction types.

Dimensional Star Schema ERD

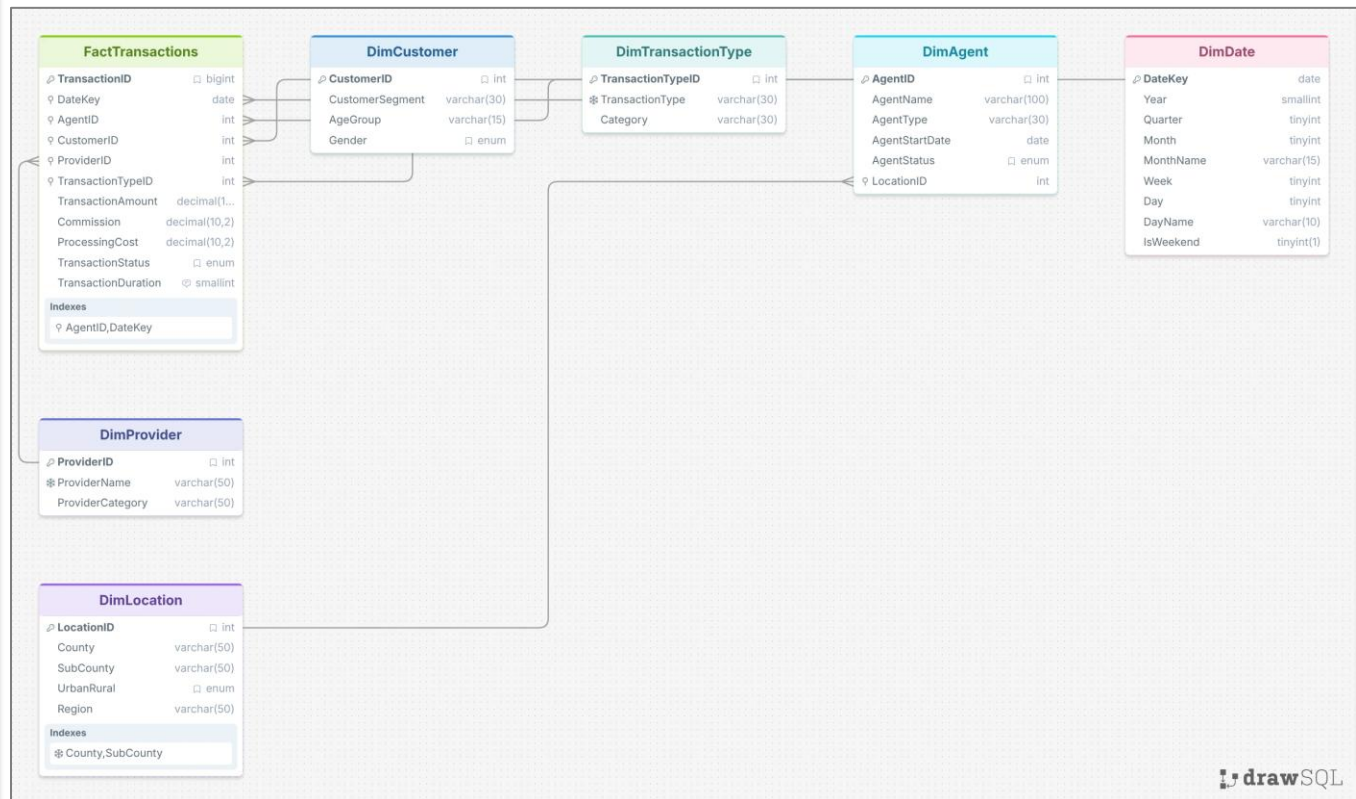


Figure 1: Dimensional data model. One fact table (*FactTransactions*) surrounded by five-dimension tables (*DimDate*, *DimAgent*, *DimCustomer*, *DimProvider*, *DimTransactionType*), with *DimLocation* linked through *DimAgent*.

2. Executive Summary

The network is **healthy and highly profitable** overall, generating an **88% net revenue margin** on KES 41.6M in gross revenue. However, digging beneath the headline numbers reveals a more nuanced picture:

- **Revenue is not evenly distributed:** a small number of agents and counties account for a disproportionate share of profit.
- **Volume and profitability are not the same thing:** some high-volume transaction types are comparatively low-margin, while lower-volume types like Bank Transfer punch well above their weight.
- **The network does not yet show clear saturation effects:** revenue per agent stays relatively stable (KES 3,500–3,900) regardless of how many agents operate in a county, suggesting there is room to keep growing without diluting agent economics.
- **Operational reliability is strong:** with a 95.98% transaction success rate across the network.

Executive Overview

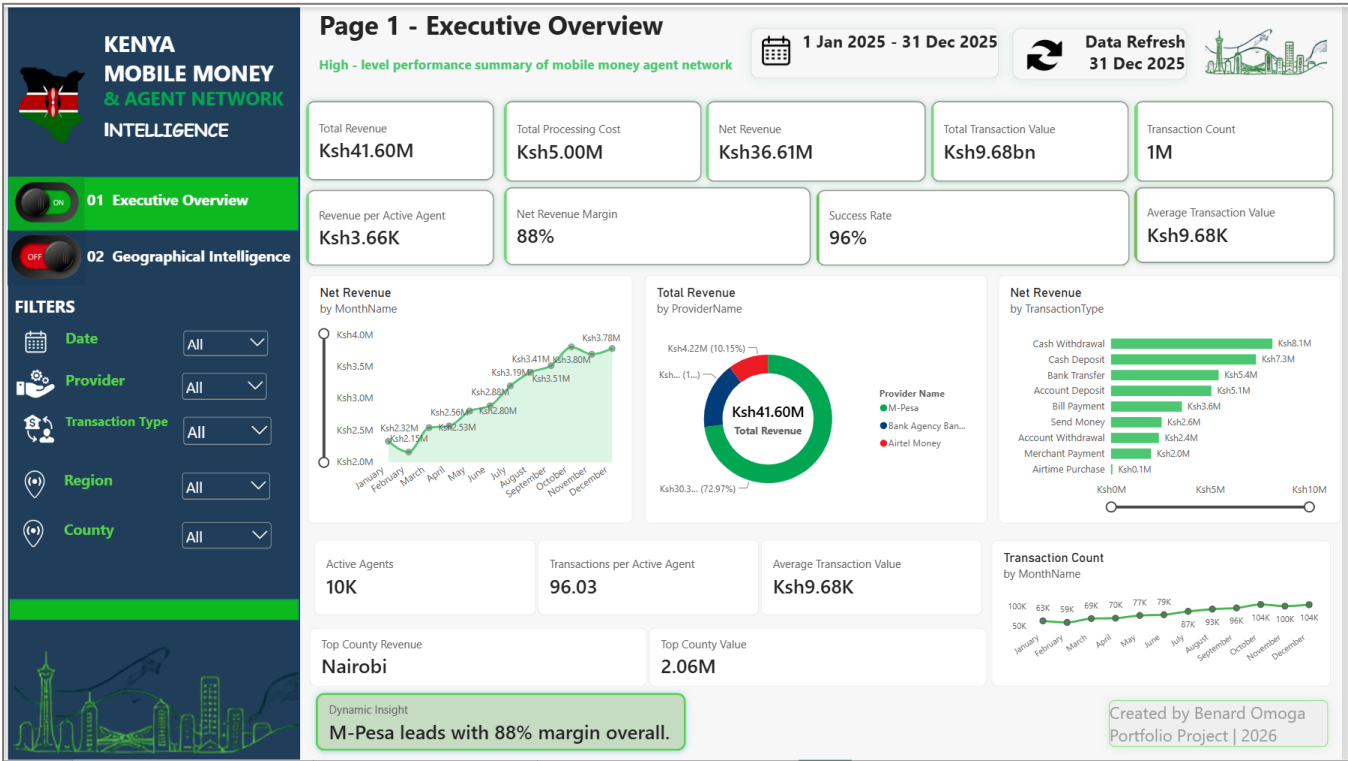


Figure 2: Executive Overview page: KPI cards, revenue trend, provider split, and transaction type ranking.

3. Key Findings

3.1 Overall Financial Health

Metric	Value
Total Transactions	1,000,000
Total Transaction Value	KES 9.68 billion
Total Revenue (Commission)	KES 41.60M
Total Processing Cost	KES 5.00M
Net Revenue	KES 36.61M
Net Revenue Margin	88%
Success Rate	95.98%
Average Transaction Value	KES 9,680

Interpretation: For every KES 100 earned in commission, KES 88 flows through to net revenue after processing costs. This is a strong, sustainable margin that gives the business room to invest in growth without compromising profitability.

3.2 Provider Performance

Provider	Share of Transactions	Approx. Revenue Share
M-Pesa	~79.5%	~73%
Airtel Money	~11%	~10%
Bank Agency Banking	~9.5%	~17%

Interpretation: M-Pesa's dominance in volume is expected and matches real-world Kenyan market dynamics. What is more interesting is that Bank Agency Banking, despite the smallest transaction volume, holds a disproportionately strong revenue share, pointing to higher-value transactions flowing through that channel (consistent with its role in larger transfers and account-based transactions).

Business implication: There is no single-provider dependency risk, the network's profitability is not hostage to one partner's performance, which is a healthy diversification signal.

Provider donut chart

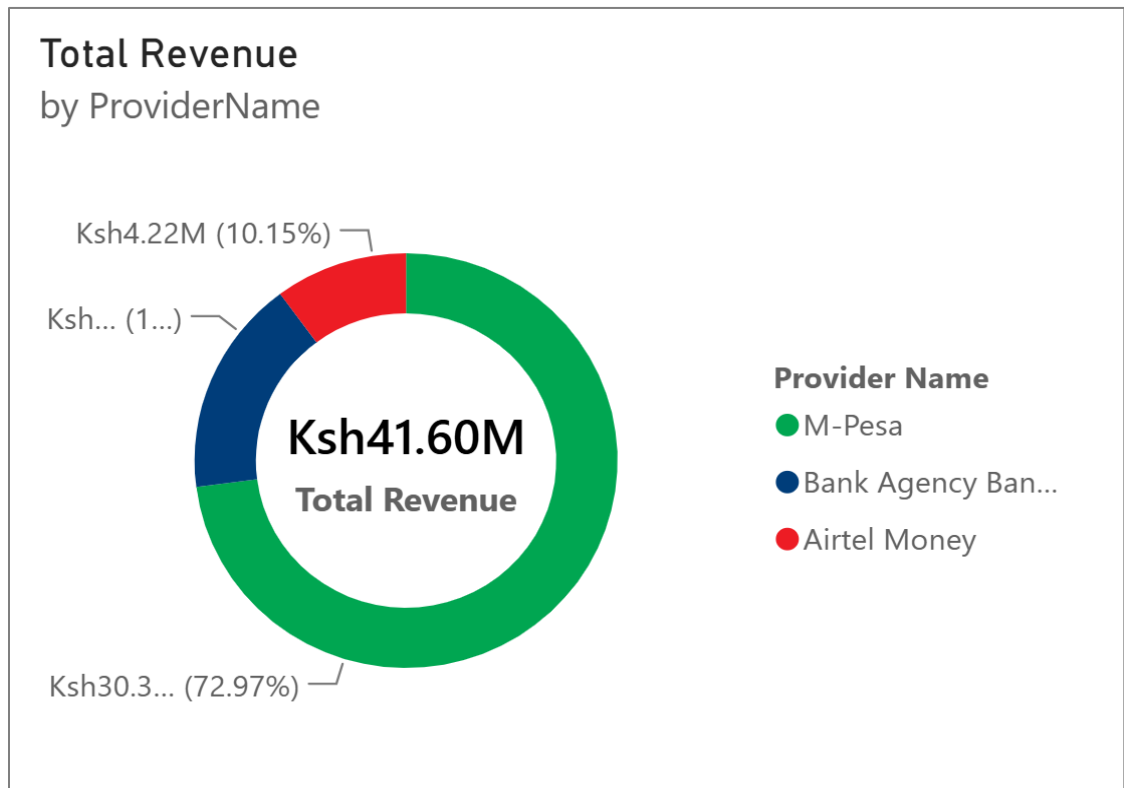


Figure 3: Revenue share by provider (M-Pesa, Airtel Money, Bank Agency Banking).

3.3 Transaction Type Profitability - Volume Is Not Profit

This was the most actionable finding in the analysis.

Transaction Type	Volume Rank	Net Revenue Rank	Insight
Cash Withdrawal	High	High	Consistent performer
Cash Deposit	High	High	Consistent performer

Send Money	High	Mid	Under-monetized relative to volume
Bank Transfer	Low	High	Disproportionately profitable per transaction
Airtime Purchase	Mid	Lowest	High volume, minimal profit contribution

Interpretation: Send Money is one of the most-used services on the platform but contributes proportionally less to net revenue than its volume would suggest. Bank Transfer, by contrast, is a lower-volume product that generates outsized profit per transaction.

Recommendation: Consider a modest commission adjustment on Send Money, and explore agent incentive programs to promote Bank Transfer, a small shift in transaction mix could meaningfully lift net revenue without requiring any growth in total volume.

Net Revenue by Transaction Type

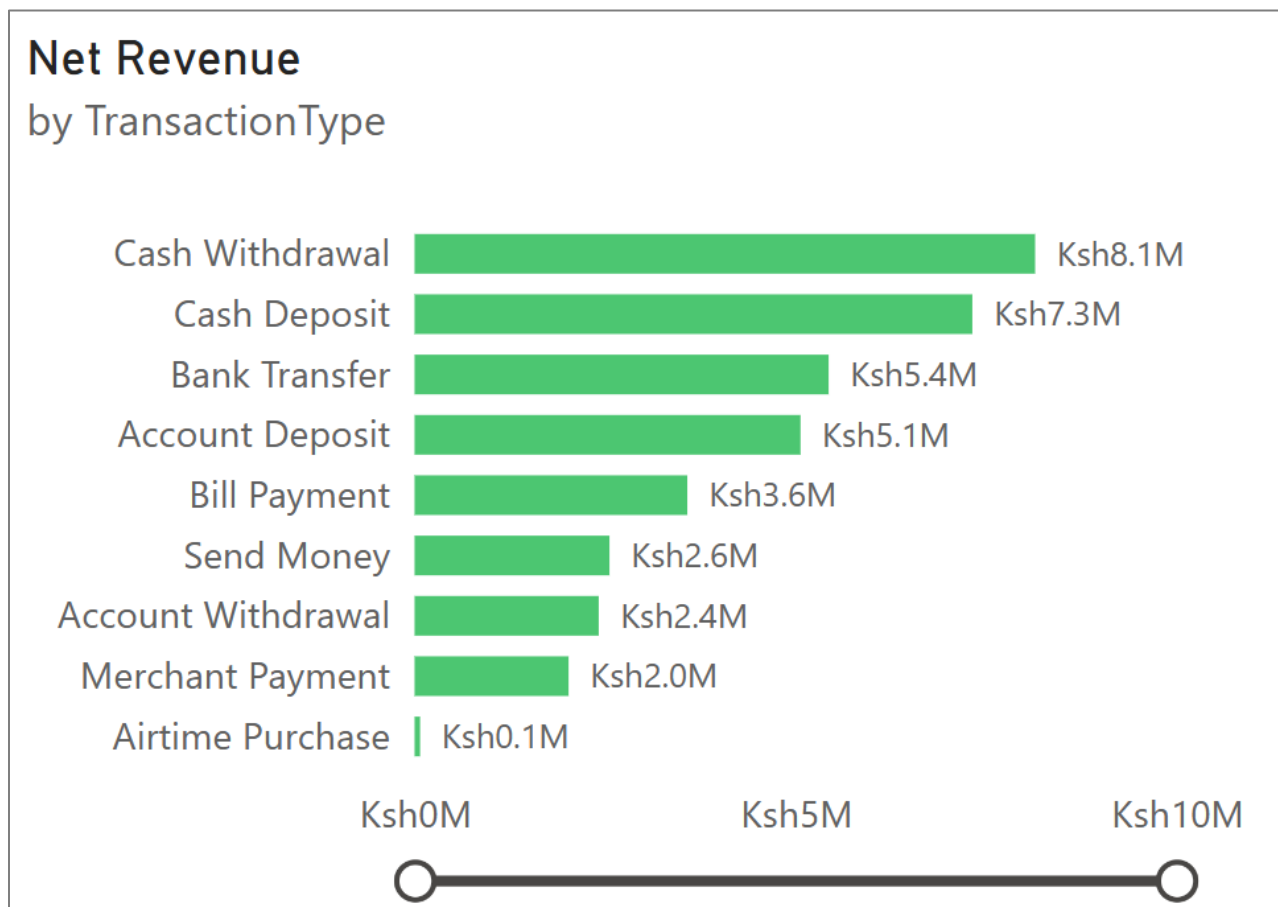


Figure 4: Net revenue ranked by transaction type; volume leaders are not always profit leaders.

3.4 Agent Performance & Revenue Concentration

The top 10 highest-earning agents (out of 10,000 total) generated roughly **KES 71,000 in combined net revenue**, individually ranging from ~KES 6,900 to ~KES 7,700, a fairly tight spread, indicating there is not a small handful of extreme outliers "super agents" dominating the network. Revenue at the top is earned by many agents performing consistently well, not a few anomalies.

Interpretation: This is a positive sign for network resilience, the business is not overly dependent on a small number of agents whose departure could meaningfully impact revenue.

Top 10 Agents table

AgentName	County	AgentType	Net Revenue
Agent 03826	Homa Bay	Merchant Agent	Ksh7,690.74
Agent 08483	Isiolo	Standard Agent	Ksh7,514.41
Agent 00753	West Pokot	Standard Agent	Ksh7,312.07
Agent 03486	Uasin Gishu	Standard Agent	Ksh7,083.29
Agent 00218	Nakuru	Standard Agent	Ksh7,046.58
Agent 09871	Mandera	Standard Agent	Ksh7,034.48
Agent 09266	Bomet	Standard Agent	Ksh6,974.06
Agent 07943	Kiambu	Standard Agent	Ksh6,944.94
Agent 06132	West Pokot	Standard Agent	Ksh6,914.02
Agent 03128	Meru	Standard Agent	Ksh6,896.15
Total			Ksh71,410.74

Figure 5: Top 10 agents by net revenue, showing county and agent type.

3.5 Geographic Performance

Geographic Intelligence

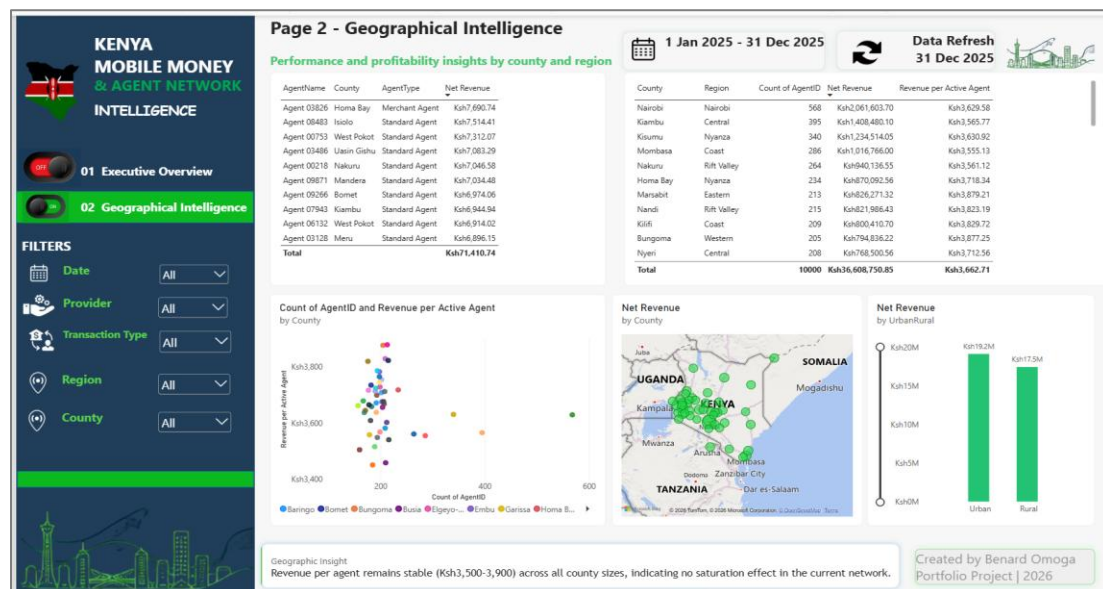


Figure 6: Geographic Intelligence page; county ranking table, scatter plot, map, and urban/rural comparison.

County	Rank	Net Revenue	Agents	Revenue per Agent
Nairobi	#1	~KES 2.06M	568	KES 3,629.58
Kiambu	#2	~KES 1.41M	395	KES 3,565.77
Kisumu	#3	~KES 1.23M	340	KES 3,630.92
Mombasa	#4	~KES 1.02M	286	KES 3,555.13

Interpretation: Nairobi leads in total revenue, as expected, it has the largest agent network by far. However, its revenue-per-agent figure (KES 3,629.58) is **not meaningfully different** from smaller counties like Kisumu (KES 3,630.92) or Mombasa (KES 3,555.13).

This is a genuinely important, and slightly counter-intuitive, finding: **the data does not show a strong urban saturation effect**. The original hypothesis (that adding more agents in dense urban markets dilutes per-agent economics) is not strongly supported by this dataset. Revenue per agent remains broadly stable (roughly KES 3,500–3,900) regardless of how many agents operate in a given county.

Business implication: Rather than treating Nairobi as "maxed out" and secondary cities as the only growth lever, the data suggests **expansion in any county appears sustainable** without materially diluting individual agent earnings. This is a more optimistic growth story than initially hypothesized, and it is worth stating plainly rather than forcing a saturation narrative the data does not fully support.

Scatter plot: Revenue per Active Agent by County

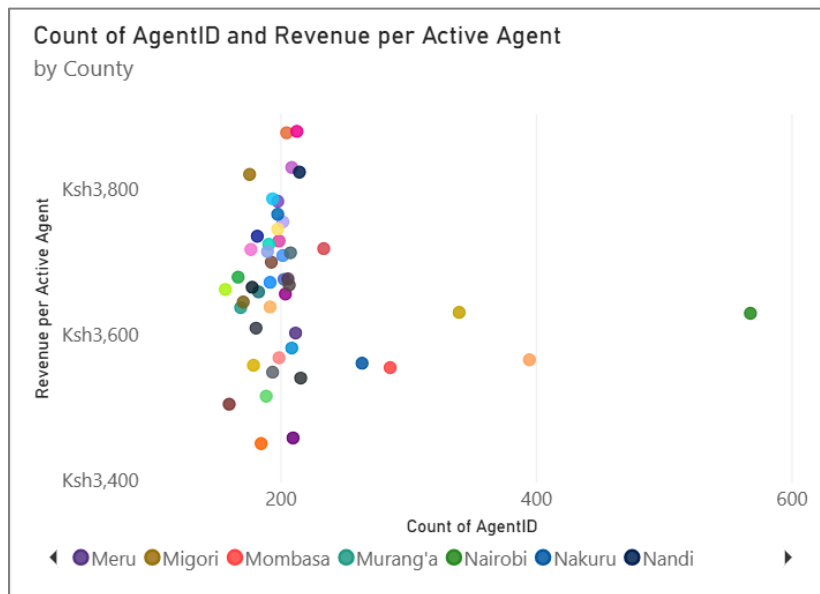


Figure 7: Agent count vs. revenue per agent by county; the flat pattern here is the evidence behind the "no strong saturation" finding.

Kenya revenue map

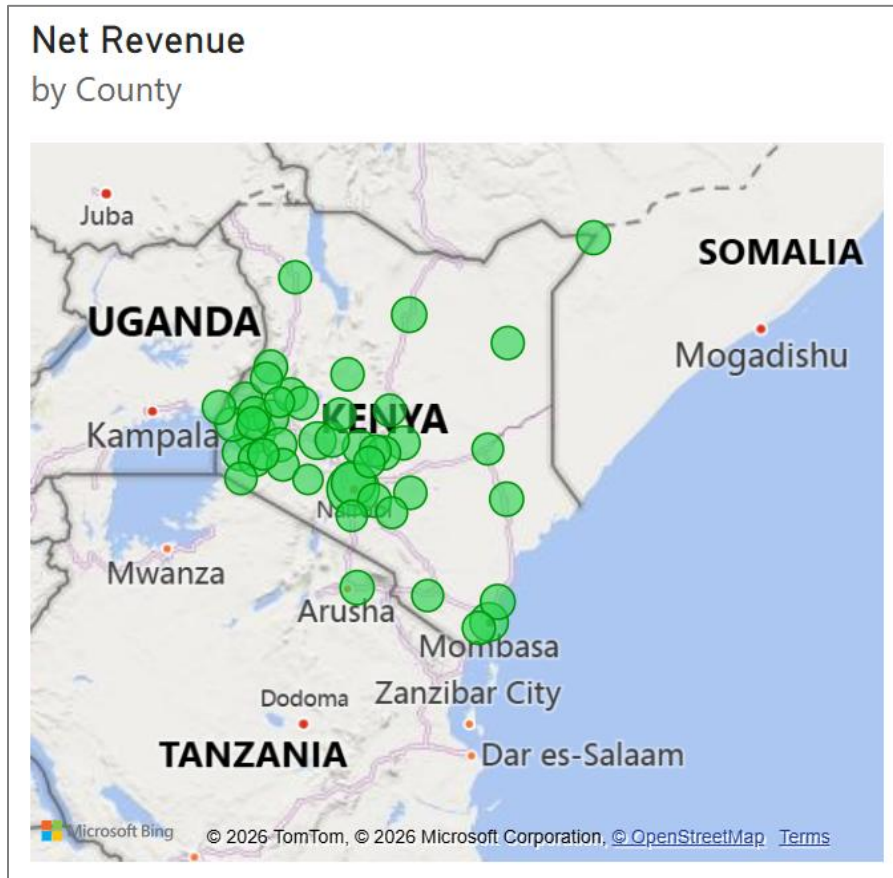


Figure 8: Net revenue by county, mapped geographically, bubble size indicates revenue.

3.6 Urban vs. Rural Split

Urban and rural counties show comparable net revenue contribution when viewed at the aggregate level, though urban markets carry higher absolute transaction volume by virtue of population density. Rural markets are not underperforming relative to their agent counts, this reinforces the geographic finding above.

3.7 Operational Reliability

- **Success rate:** 95.98%
- **Failed transactions:** ~3.02%
- **Reversed transactions:** ~0.99%

These figures fall within a healthy, realistic range for a mobile money platform of this scale, with no single provider or transaction type showing an alarming failure pattern.

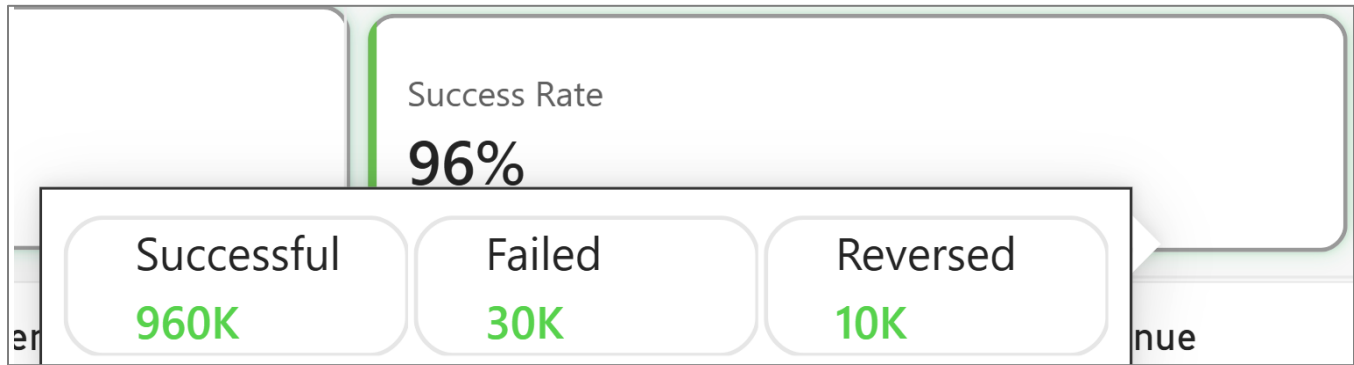
Success Rate tooltip

Figure 9: Interactive tooltip on the Success Rate KPI card, showing the Success/Failed/Reversed breakdown.

4. Strategic Recommendations

Recommendation 1: Rebalance Commission Structure Toward Bank Transfer

Bank Transfer generates outsized profit per transaction relative to its volume. Introducing agent incentives or lightly repricing Send Money could shift transaction mix toward higher-margin products without requiring new customer acquisition.

Expected impact: Incremental net revenue lift, achievable through pricing/incentive changes alone.

Recommendation 2: Pursue Balanced Geographic Expansion

Since revenue-per-agent remains stable across county sizes, expansion should not be limited to "escaping saturated Nairobi." Growth can be pursued broadly, in Nairobi, secondary cities, and rural counties alike, based on market opportunity rather than a false assumption of urban dilution.

Expected impact: More flexible, opportunity-driven expansion strategy rather than an artificially constrained one.

Recommendation 3: Protect Agent Network Diversity

With no small group of "super agents" dominating revenue, the network shows healthy resilience. Continued investment in agent training and support across the broad base, rather than concentrating resources on a few top performers, will help sustain this stability.

Recommendation 4: Monitor Cash Withdrawal Success Rates

While overall reliability is strong, Cash Withdrawal showed a marginally elevated failure pattern in exploratory analysis. A closer operational review (network latency, PIN entry friction, agent float availability) could recover a meaningful number of currently-failed transactions.

5. How the Dashboard Answers the Business Question

Phase 1 Question	Dashboard Component
How profitable is the network?	Executive Overview KPIs (Net Revenue, Margin)
Where are we performing well?	Provider donut, Transaction Type ranking, County table
Where are we losing money or facing risk?	Success Rate tooltip breakdown, Transaction Type comparison
Does growth translate to profitability?	Geographic scatter plot (Agent Count vs. Revenue per Agent)
Where should we expand?	Geographic Intelligence map + county ranking table

6. Methodology Note

This project used **synthetic data** generated via a Python script with realistic statistical distributions (log-normal transaction amounts, weighted provider/transaction-type probabilities grounded in Communications Authority of Kenya market share data). While the underlying transactions are not real, the modeling choices were designed to produce patterns consistent with how a real Kenyan mobile money network behaves, which is why findings like "no strong saturation effect" are reported honestly rather than adjusted to fit a predetermined narrative.

Where the data showed a different story than the original hypothesis (e.g., Section 3.5), that has been reported as-is. This reflects the same discipline a real analyst would apply: let the data speak, even when it complicates the initial thesis.

7. Project Deliverables

- MySQL star schema (7 tables, 1M+ fact rows)
- Python synthetic data generation pipeline
- 10 SQL exploratory analysis queries with documented findings
- Power BI dashboard (2 pages + tooltip page):
 - Executive Overview
 - Geographic Intelligence
- 9+ DAX measures
- This business insights write-up